

Recommended PET Configuration

Comparison with Gregor's Implementation

Publication Agent B (PET)

18 September 2026

PET is diagnostic method development, not a publication product.
Nothing here is a publication adoption, an uncertainty, or a Gate-6 action.

Keep our estimator's structure. Borrow five things from Gregor. Test two of them before anything else.

Keep (evidence supports, or load-bearing for OmniFold)

- ▶ muon as a distinguished FiLM event input
- ▶ frozen MC normalization; periodic azimuth
- ▶ family pooling as the practical default
- ▶ energy-ranked retention
- ▶ KNN locality in detector coordinates
- ▶ the whole OmniFold construction

Borrow (candidates — none deployed)

- ▶ **per-family log energy sums**, computed over the whole event
- ▶ **aggregate overflow**, energy-weighted merge
- ▶ richer per-token features + type embedding
- ▶ warmup + cosine decay, gradient clipping
- ▶ one capacity step

Do first, zero new compute: measure whether typed objects cover the cluster energy. That one number gates the three largest borrowings.

Do not borrow: muon-as-token; zero-fill for absent fields; embedding index 0 shared with padding; input LayerNorm; raw ϕ ; concatenation-order truncation; weight decay.

What is pinned, and what “ours” means

label	what it is	pin
ours, production	pet-fullevent-fps-v1: the estimator the driver stamps	44142e8c (branch pet-direct-token-comparison)
ours, experimental	four-arm representation study: typed descriptors, routing, aggregate overflow. No arm is in the production path.	44142e8c, direct_token_comparison/
ours, proposed	what this deck recommends changing	nothing is built
Gregor's	gregorkrz/minerva-ml	fc9a099d, 2026-07-20, clean tree

- ▶ **fc9a099 is the repo as cloned, not a paper commit.** No release tags; the repo has evolved past arXiv:2604.12364. The paper commit must come from Gregor.
- ▶ Everything attributed to him is the **default** unless marked optional. Verified at fc9a099: use_pid on, use_cond on, include-E-sum on, max_particles 33, **use-max-blobs-and-prongs off** — so his overflow aggregation is optional and not the default.
- ▶ **He is not doing unfolding.** Supervised regression/classification, MC only, no data leg, no per-event weights. Several rows below have no counterpart, and that is a difference in problem, not in quality.

The actual baseline: our production estimator

Data path

- ▶ cloud = **raw non-muon calorimeter clusters** ($E, \text{pos}, z, \text{view}, t$), 5 columns
- ▶ muon **removed from the cloud**; carried as a 13-feature event block, FiLM conditioning
- ▶ cap: stable energy sort, **top 12**, zero-pad — applied **at dump time**, so changing it needs a re-dump
- ▶ overflow **discarded with no trace**: no count, no summed energy, no flag
- ▶ fixed physical divisors; event features z-normalized with *frozen* reco-MC pass_reco statistics
- ▶ azimuth as $(\cos \phi, \sin \phi)$ everywhere

Model and training

- ▶ PET: encoder $\rightarrow$ **KNN local aggregation** ($K = 3$ over (pos, z)) $\rightarrow$ 2 blocks, 2 heads, width 32, LayerScale $\rightarrow$ class token + FiLM head
- ▶ **47,041 trainable parameters**; 93,954 per OmniFold iteration
- ▶ weighted BCE; Adam 10^{-4} annealed to 10^{-5} , realized LR *asserted* at runtime; batch 512; niter 3, epochs 8; 2 M subsample
- ▶ MultiFold; push weight $w = \exp(\text{logit})$ in logit space
- ▶ background: negweight-refined, fail-closed
- ▶ seed 42 frozen — this is what makes C_{stat} statistical

`train_fullevent_nominal.py:403-407, 47-71; fullevent_fps_dataloader.py; dump_pointcloud_inputs.py:90-112; omnifold_nn/omnifold/{net,omnifold}.py; receipts/model-capacity.json`

Data path

- ▶ cloud = **reconstructed objects**: muon, ≤ 2 photons, blobs, prongs — 10 columns each
- ▶ η , ϕ , $\log p_T$, $\log E$, **PID code** (8-class learned embedding), dE/dx , x, y, z, t (all $/10^4$)
- ▶ muon is **a token like any other** (PID 0)
- ▶ caps in series: 150 at preprocessing (energy-sorted *only* if exceeded), then **first 33 rows at batch time, no re-sort**
- ▶ absent fields **zero-filled, no validity mask**
- ▶ 16 global features: recoil calorimetry, passive material, Michel, $\gamma\gamma$ mass, prong count, **6 per-PID log energy sums**

Model and training

- ▶ PointGlobalMixedViT: mixed encoder $\rightarrow$ **Fourier positional MLP over** $(\eta, \phi) \rightarrow 4$ pre-norm blocks, 8 heads, $d_{\text{model}} = 128 \rightarrow$ CLS fused with EVT token
- ▶ **890,130 parameters** — $18.9\times$ ours
- ▶ Huber on $\log 1p$; **AdamW**, wd 0.01, **1000-step warmup then cosine, grad clip 1.0**; batch 2048; 100k–500k steps
- ▶ optional: OmniLearned PET2, HyperScale, BERT, pretrained init
- ▶ **pretrained checkpoints are unreachable** — HyperScale repo 404, NERSC URLs time out (audit B_provenance)

vit.py:226-373; train.py:465-756, 2328-2358; preprocessing.py:257-583; jobs/submit_train_jobs.py:150-151;
gregor-audit/jobs/A_dataset.out.md

Comparison I — representation

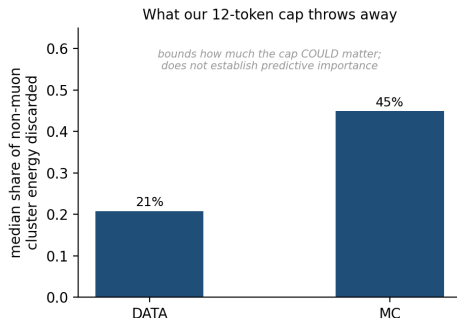
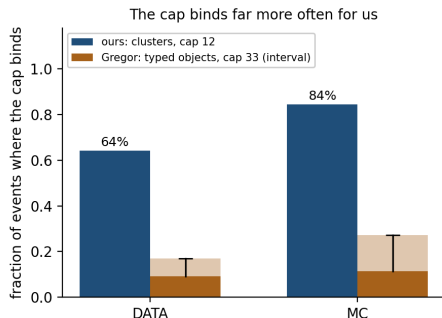
	ours	Gregor's	consequence	recommendation
objects	raw non-muon calorimeter clusters; muon distinguished	reconstructed objects; muon is a token	mean 74.6/102.6 clusters vs 14.0/15.6 objects: $5.3 \times / 6.6 \times$ fewer tokens	borrow the vocabulary; keep the distinguished muon
token features	5 cols, no type code; mask = ($E=0$)	10 cols incl. direction, dE/dx , time, PID embedding	more physics per object; but zero-fill hides absence, and padding shares PID index 0 with muon (masked, so latent)	borrow features + embedding; reserve index 0 for padding
transforms	physical divisors; frozen MC z-norm; $(\cos \phi, \sin \phi)$	logs $+ / 10^4$; LayerNorm inside encoder; raw ϕ	raw ϕ is discontinuous at the wrap — we fixed this ourselves in July	keep ours; recommend the ϕ fix to him
event globals	13, all muon/vertex	16, recoil calorimetry + 6 per-PID energy sums	his sums are a fixed-width channel that survives a cap — but he computes them <i>after</i> truncation	borrow the sums, computed <i>before</i> any cap
caps and overflow	top 12 by energy, rest discarded silently	150 then first 33 unsorted; aggregate token <i>optional</i> , <i>off</i>	our cap binds 64%/84% vs his 9–17%/11–27%; his ordering is worse	borrow aggregation; keep energy ranking

Comparison II — model, training, integration

	ours	Gregor's	consequence	recommendation
routing	one family, so no question in production; pooled vs individual measured experimentally	individual, unconditionally	no accuracy difference detected; individual costs $1.24\times$ at the operating point, $1.88\times$ in the tail	keep pooling as a practical default
architecture	2 blocks, 2 heads, width 32, KNN in (pos, z) ; 47k params	4 blocks, 8 heads, width 128, Fourier PE in (η, ϕ) ; 890k params	$18.9\times$ gap; ours is retrained ~ 130 times per campaign, his once	keep architecture; treat capacity as open
init	random, seed 42 frozen (this makes C_{stat} statistical)	random; optional pretrained OmniLearned	checkpoints unreachable : 404 and timeouts	keep; ask Gregor for the weights and hashes
training	Adam $10^{-4} \rightarrow 10^{-5}$ step anneal; no wd, no clip	AdamW, wd 0.01, warmup + cosine, clip 1.0	his schedule is better conditioned and free; wd biases a likelihood-ratio classifier ($w = e^{\text{logit}}$)	borrow warmup/cosine/clip; not weight decay
OmniFold	full MultiFold, logit-space ratio, negweight background, fail-closed data path; closures PASS	no counterpart — MC only, no weights, no flags	his <i>dataset</i> cannot be an OmniFold input; his <i>vocabulary</i> can	keep ours in full

Before any numeric comparison with his results: his E_{avail} uses full charged-pion energy + $K^\pm$ against our pion-KE definition — an offset of order 140 MeV per $\pi^\pm$.

The measured problem: our cap binds, and it binds hard



Unselected, unweighted tuple entries: 17,930 data (1B) + 186,439 MC (1A). Not the selected population.

Same two tuple files, both vocabularies. Our 12-token cluster cap binds in **64.1 % (data) / 84.4 % (MC)** of events and discards a median **20.8 % / 44.9 %** of non-muon cluster energy. A 33-token typed-object cap binds in **9.2–16.9 % / 11.4–27.2 %** — an interval, because the receipt holds marginal, not joint, family multiplicities.

Three things this does not show. It does not show the discarded clusters carry predictive information. It does not predict a closure improvement — **no closure measurement of aggregation exists**. And these are **unselected, unweighted** tuple entries, so they do not determine selected-population behaviour.

`receipts/typed-object-budget.json; source-multiplicity.json (job 58470099)`

Measured accuracy and stability: routing is inconclusive

Eight paired seeds, four-object synthetic fixture,
information content / model size / budget / init / seeds
held identical between arms.

paired improvement, individual over pooled	
median	+9.7 %
mean	+0.8 %
seed-to-seed sd	25.8 points
95 % interval on the mean	$[-20.7, +22.4]$ %
paired p	0.50
favourable seeds	6 of 8

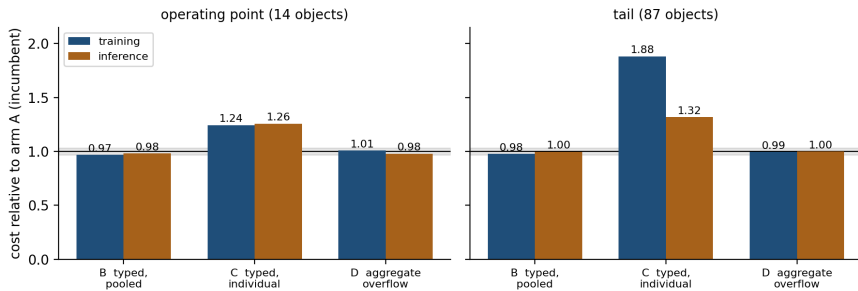
Frozen decision: NO_PASS, 142 of 146 checks holding.
Two failures *were* the inconclusive result; separately
shuffle-71 missed two projection checks marginally
(0.0119 vs 0.01; absolute errors well inside 0.05).

What each number measures —

- ▶ the interval is on the **mean paired injected-improvement percentage across training seeds on a synthetic fixture**. Not on production closure, not on real data.
- ▶ $p = 0.50$ is a paired test **across training seeds**. It measures seed scatter. **Test-sample uncertainty was not separately estimated**.
- ▶ 25.8 points is a **planning assumption** carried from a different fixture, two arms and a different denominator — **not measured power for the endpoint now specified**.

And: in the fixture every generic slot is noise while in production the cluster cloud carries most of the information. Neither direction nor magnitude of a synthetic effect transfers automatically.

Measured cost: what each change actually buys and spends



Grey band: $\pm 3.3\%$, the resolution of this measurement, estimated from the arm-A/arm-B null contrast. A bar inside it is not a difference.

One A100, TF32 off, determinism on, float32. Cost is cost: a cheaper arm is not a better one.

Aggregate overflow is *not* free — it is indistinguishable from parity at this measurement's resolution. Arm B is computationally identical to arm A in the pooled configuration and still reads $0.97\times$, so the floor is about $\pm 3.3\%$. The non-compute cost is real: a dump regeneration, a schema field, a re-validation, and a new estimator fingerprint.

Individual routing is outside that band and grows with multiplicity ($1.24\times \rightarrow 1.88\times$ training). Pooling is nearly flat. Our campaign multiplier is ~ 130 trains, so per-train cost is multiplied, not absorbed.

`cost-half.json` (one A100, TF32 off, determinism on, float32); `inference-benchmark.json`

borrow (candidate)	why	measured cost	prerequisite
per-family log energy sums, over the <i>whole</i> event	cap-proof fixed-width channel	≈ 0	fingerprint change; rank alongside eavail/q3
aggregate overflow, energy-weighted merge	addresses a measured 21–45 % energy loss in most events	parity to $\pm 3.3\%$	dump regeneration; no closure evidence yet
richer per-token features + type embedding, index 0 reserved	more physics per token, fewer tokens	unmeasured	typed-object energy coverage
warmup + cosine, gradient clipping	better-conditioned schedule	≈ 0	Gate-4 fingerprint; closure re-run
one capacity step	18.9× gap is a live hypothesis	multiplies through the UQ campaign	price against the campaign, not one train

Deferred, deliberately: individual-object routing — unresolved on accuracy and measurably more expensive.

Decided without new work: normalization, pretraining availability, OmniFold integration, numerical behaviour.

Back to Gregor: encode ϕ periodically; reserve embedding index 0 for padding; map raw prong PID 9 (it currently raises); compute the per-PID energy sums before truncation.

Limitations — what these numbers do not say

1. “Negligible measured overhead” is not “free”. Aggregate overflow is inside the $\pm 3.3\%$ resolution of the cost measurement, and it carries a dump regeneration, a schema change, a re-validation and a new fingerprint.
2. Discarded energy does not establish closure improvement. 21–45 % bounds how much the cap *could* matter. No closure measurement of aggregation exists on either side.
3. Validation is incomplete: 10 of 13 widths released. Three widths failed the finite-difference check V5 and were not released. Diagnosis: the *criterion* was wrong, not the gradient — error falls as h^2 (1.8×10^{-2} at the frozen $h=10^{-2}$, 3.1×10^{-4} at 10^{-3}) then rises, the signature of truncation giving way to cancellation. **The criterion was not relaxed after the failures.**
4. Unselected source measurements do not determine selected behaviour. Two files, unweighted, no production selection, no POT weighting.
5. Every statistical quantity is labelled by what it measures — seed scatter, not test-sample uncertainty; a synthetic fixture, not production; a planning assumption, not measured power.
6. The pin is the repo, not the paper. fc9a099 may not be what produced arXiv:2604.12364.

One question, no compute, gates the three largest borrowings:

Do blobs + prongs + photons cover the non-muon cluster energy on our tuples?

A counts-and-energies read of branches the A1 authorization already covers; order one CPU core-hour. If coverage is poor, the typed vocabulary is dead and the list collapses to aggregate overflow alone.

Then one bounded proposal, not five experiments. Aggregation, energy sums, routing and capacity *interact*: energy sums partly substitute for aggregation; routing matters more when more objects survive the cap; and a 47k-parameter model may have no room for a richer representation, which would make every single-factor test read “no effect” for the wrong reason. The right instrument is a **matched complete-configuration comparison** — production against one assembled candidate on the same endpoint and seeds — with single-factor arms added only where it shows an effect.

Already authorized and currently blocked. The variance pilot (≤ 30 GPU-h) would measure whether any of this is resolvable inside the 130 authorized. Its precondition is complete validation, which is unmet. **One decision unblocks it:** amend the frozen V5 criterion to a per-coordinate step rule. That is Joseph’s call, and I did not make it.

Campaign spend to date: **16.0 of 290 GPU device-hours**, 2.0 CPU core-hours.

Appendix A — claim-to-evidence index

claim	evidence
cap binds 64.1 % / 84.4 %; median 20.8 % / 44.9 % energy discarded	direct_token_comparison/local_validation/20260918-source/source-multiplicity.json (job 58470099; 17,930 data + 186,439 MC entries)
typed cap binds 9.2–16.9 % / 11.4–27.2 %; 5.3×/6.6× fewer tokens	receipts/typed-object-budget.json (derived; 8 tests)
47,041 vs 890,130 parameters	receipts/model-capacity.json (both models instantiated)
routing: median +9.7 %, [−20.7, +22.4] %, $p = 0.50$, NO_PASS 142/146	local_validation/20260917-matrix/matrix-summary.json
inference 1.283×, all 5 checks pass, $CV \leq 2\%$	local_validation/20260917-inference/inference-benchmark.json
GPU cost by arm; A/D parity; $\pm 3.3\%$ floor	local_validation/20260918-a3-cost/cost-half.json
10 of 13 widths released, no hard stop	local_validation/20260918-tailval/validation-final.json; criteria sha256 abccd88b...
Gregor token format, PID scheme, caps, globals	gregor-audit/jobs/A_dataset.out.md (line-cited to fc9a099)
checkpoints unreachable; OmniLearned MIT;	gregor-audit/jobs/B_provenance.out.md
dataset CC-BY-4.0	
our production configuration	FULL_EVENT_FEATURE_CONTRACT.md; train_fullevent_nominal.py:47–71, 403–407

Full index: ../CLAIM_EVIDENCE_INDEX-20260918.md. Full inventory: ../CONFIGURATION_COMPARISON-20260918.md.

Appendix B — the V5 diagnosis, in full

Three widths — (0, 64, 2) and (2, 160, 12) for the individual arm, (2, 12, 4) for both — failed the finite-difference check, and only that check. My first hypothesis (a small denominator) was **wrong**: the failing coordinates have $|\nabla| \approx 0.0215$. Sweeping the step at the worst coordinate:

	h	numeric derivative	relative error
	3×10^{-2}	-0.01787	1.69×10^{-1}
1×10^{-2} (<i>frozen</i>)		-0.02111	1.84×10^{-2}
	3×10^{-3}	-0.02147	1.67×10^{-3}
	1×10^{-3}	-0.021497	3.14×10^{-4}
	3×10^{-4}	-0.021531	1.30×10^{-3}

Error falls as h^2 and then rises: truncation giving way to cancellation. The analytic gradient is correct; the frozen *global* step was too large at those coordinates, and the optimal step is per-coordinate. The plateau check nearly caught it (16.8% spread against a 20% tolerance).

I did not relax the criterion. Releasing those widths requires an amended step rule. Until then the execution path is validated for 10 of 13 widths, which is **incomplete**, and the variance pilot's precondition is unmet.

Appendix C — what the campaign cost, and what it did not authorize

item	GPU device-hours	source
representation matrix (24 paired jobs)	13.27	matrix-summary.json
A3 geometry probe + width gate (5 subs)	0.43	GO_NO_GO-20260918.md §7
tail validation (4 of 4 submissions)	0.77 of 1.5 authorized	validation-final.json
inference benchmark + matrix closeout	<i>not itemized here</i>	—
A1 source characterization	(CPU: 2.0 core-h)	source-multiplicity.json
campaign total	16.0 of 290	RECOMMENDATION-20260918.md

The first three rows are itemized from their own receipts; the total is carried from the campaign recommendation. They do not sum, because the inference benchmark (including one 35-minute timeout that measured nothing) and the matrix closeout are inside the total but are not separately itemized in any receipt I can cite. Stated as a gap rather than closed by arithmetic.

Not launched: the variance pilot (≤ 30 GPU-h, precondition unmet) and Stage 2 (≤ 130 GPU-h, downstream of the pilot).

Not authorized by anything in this deck: publication adoption, any covariance or systematic, a central-value change, a Gate-6 action, or the construction of C_{ML} . Typed objects and aggregate overflow remain *prospective* representation improvements, exactly as the analysis note already says.

Not done: nothing in this deck has been sent to anyone.